

**BHAGALPUR, India**WMO: **424980**Lat: **25.233N**Lon: **86.950E**Elev: **161**StdP: **14.61**Time Zone: **5.50 (IND)**Period: **01-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |             |             |             |             |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|-------------|-------------|-------------|-------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |             |             | 99%         |             |             | 0.4%                  |             | 1%         |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR          | MCDB        | DP          | HR          | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>50.9 | (c)<br>53.2 | (d)<br>46.9                   | (e)<br>47.8 | (f)<br>56.7 | (g)<br>48.6 | (h)<br>51.0 | (i)<br>58.4 | (j)<br>5.5            | (k)<br>64.3 | (l)<br>5.2 | (m)<br>63.9 | (n)<br>0.9            | (o)<br>N/A | (p)<br>0.295 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>6      | (b)<br>11.8            | (c)<br>101.6    | (d)<br>81.1 | (e)<br>99.6 | (f)<br>81.0 | (g)<br>97.7 | (h)<br>81.2 | (i)<br>86.6         | (j)<br>95.7 | (k)<br>85.4 | (l)<br>93.8 | (m)<br>84.8 | (n)<br>92.7 | (o)<br>1.9           | (p)<br>N/A |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 84.5                            | 182.0 | 91.8 | 83.5 | 176.2 | 90.4 | 82.8 | 172.1 | 89.6 | 51.8          | 95.6 | 50.2 | 94.7 | 49.5 | 93.8 | 92.1              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |           |          |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|-----------|----------|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |           |          |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      |           |          |
| (a) 5.5           | (b) 5.0 | (c) 4.1 | (d) DB                     | (e) 47.1 | (f) 104.8          | (g) 2.1 | (h) 3.7                                            | (i) 45.6 | (j) 107.4  | (k) 44.3 | (l) 109.6  | (m) 43.1 | (n) 111.7  | (o) 41.6 | (p) 114.4 | (q) 94.3 |
|                   |         |         | (d) WB                     | (e) 47.1 | (f) 87.7           | (g) 2.6 | (h) 2.6                                            | (i) 45.2 | (j) 89.6   | (k) 43.7 | (l) 91.1   | (m) 42.2 | (n) 92.5   | (o) 40.3 | (p) 94.3  |          |

**Monthly Climatic Design Conditions**

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 79.3   | 61.2  | 69.3  | 79.5  | 86.2  | 88.0  | 88.9  | 86.0  | 86.6  | 85.8  | 81.9  | 73.9  | 64.1  |
|                                                                        | DBStd       | 10.12  | 4.14  | 4.48  | 4.34  | 4.13  | 4.20  | 3.83  | 2.91  | 2.86  | 3.12  | 3.69  | 3.93  | 4.37  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 205    | 125   | 11    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 69    |
|                                                                        | CDD50       | 10702  | 348   | 541   | 915   | 1086  | 1178  | 1167  | 1117  | 1133  | 1074  | 989   | 716   | 438   |
|                                                                        | CDD65       | 5434   | 8     | 133   | 450   | 636   | 713   | 717   | 652   | 668   | 624   | 524   | 266   | 43    |
|                                                                        | CDH74       | 69208  | 15    | 654   | 4735  | 9049  | 10467 | 10673 | 8789  | 9134  | 8361  | 5723  | 1512  | 96    |
| (13)                                                                   | CDH80       | 35441  | 0     | 122   | 1975  | 5152  | 6177  | 6392  | 4354  | 4683  | 4150  | 2140  | 287   | 9     |
| (14)                                                                   | Wind        | WSAvg  | 0.9   | 0.7   | 0.9   | 1.2   | 1.7   | 1.6   | 1.2   | 0.8   | 0.8   | 0.5   | 0.3   | 0.5   |
| (15) Precipitation                                                     | PrecAvg     | 45.8   | 0.5   | 0.5   | 0.5   | 0.9   | 2.6   | 5.9   | 12.2  | 10.4  | 8.6   | 2.8   | 0.4   | 0.2   |
|                                                                        | PrecMax     | 75.0   | 1.6   | 1.5   | 2.4   | 3.0   | 9.1   | 13.0  | 19.4  | 17.6  | 24.3  | 12.1  | 4.0   | 2.0   |
|                                                                        | PrecMin     | 28.5   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.3   | 6.3   | 4.3   | 2.2   | 0.1   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 10.7   | 0.5   | 0.4   | 0.6   | 1.0   | 2.3   | 3.7   | 3.9   | 4.0   | 5.6   | 3.0   | 0.8   | 0.5   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 75.8  | 86.9  | 97.1  | 102.1 | 104.8 | 104.1 | 98.0  | 97.9  | 96.6  | 95.3  | 88.7  | 81.3  |
|                                                                        |             | MCWB   | 66.1  | 72.0  | 74.9  | 77.6  | 79.4  | 83.4  | 85.3  | 85.8  | 85.7  | 82.8  | 75.4  | 71.8  |
|                                                                        | 2%          | DB     | 72.6  | 83.6  | 94.5  | 99.7  | 102.2 | 101.2 | 95.1  | 95.5  | 94.4  | 92.2  | 85.8  | 76.7  |
|                                                                        |             | MCWB   | 64.1  | 69.6  | 74.5  | 76.0  | 79.6  | 84.4  | 84.2  | 84.8  | 84.5  | 81.4  | 74.5  | 67.1  |
|                                                                        | 5%          | DB     | 70.3  | 80.5  | 91.7  | 98.0  | 99.7  | 99.3  | 93.0  | 93.4  | 92.8  | 90.0  | 83.1  | 73.6  |
|                                                                        |             | MCWB   | 63.6  | 68.2  | 73.6  | 76.8  | 80.6  | 84.1  | 83.6  | 84.0  | 83.6  | 80.7  | 74.1  | 66.6  |
|                                                                        | 10%         | DB     | 68.4  | 77.8  | 89.0  | 95.9  | 97.2  | 97.0  | 91.1  | 91.6  | 91.3  | 88.2  | 80.9  | 71.2  |
|                                                                        |             | MCWB   | 62.8  | 67.0  | 73.2  | 77.5  | 81.2  | 83.9  | 83.1  | 83.5  | 82.9  | 80.1  | 73.2  | 65.1  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 68.1  | 74.4  | 81.1  | 86.3  | 87.9  | 90.0  | 86.2  | 86.4  | 86.4  | 84.5  | 77.8  | 72.7  |
|                                                                        |             | MCDB   | 72.9  | 84.3  | 90.8  | 97.1  | 97.9  | 99.7  | 95.2  | 95.0  | 94.6  | 92.2  | 85.8  | 79.2  |
|                                                                        | 2%          | WB     | 66.0  | 71.7  | 78.3  | 83.7  | 85.2  | 87.3  | 85.0  | 85.3  | 85.2  | 83.0  | 76.1  | 69.0  |
|                                                                        |             | MCDB   | 70.2  | 79.9  | 87.7  | 94.7  | 95.5  | 96.4  | 92.5  | 92.7  | 92.5  | 89.6  | 83.3  | 74.0  |
|                                                                        | 5%          | WB     | 64.6  | 69.9  | 76.7  | 82.1  | 84.0  | 85.4  | 84.4  | 84.6  | 84.4  | 81.9  | 75.0  | 66.9  |
|                                                                        |             | MCDB   | 69.0  | 77.9  | 86.1  | 92.9  | 94.2  | 94.3  | 90.8  | 91.2  | 91.0  | 88.1  | 81.8  | 72.1  |
|                                                                        | 10%         | WB     | 63.3  | 68.4  | 75.2  | 80.6  | 82.9  | 84.8  | 83.7  | 84.1  | 83.6  | 80.9  | 73.8  | 65.5  |
|                                                                        |             | MCDB   | 67.7  | 76.0  | 85.2  | 90.8  | 92.6  | 93.7  | 89.5  | 90.1  | 89.7  | 86.8  | 79.8  | 70.6  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.9  | 13.6  | 16.3  | 16.1  | 14.5  | 11.8  | 8.4   | 8.2   | 8.6   | 10.4  | 11.8  | 10.9  |
|                                                                        |             | MCDBR  | 16.5  | 19.6  | 21.9  | 19.1  | 19.0  | 16.4  | 11.8  | 11.8  | 11.8  | 14.0  | 15.7  | 15.4  |
|                                                                        |             | MCWBR  | 10.1  | 10.1  | 10.2  | 8.7   | 7.7   | 6.5   | 4.6   | 4.4   | 4.9   | 5.9   | 7.0   | 8.6   |
|                                                                        | 5% WB       | MCDBR  | 13.5  | 17.5  | 18.4  | 17.3  | 16.7  | 14.1  | 10.0  | 10.2  | 10.7  | 11.3  | 13.0  | 13.6  |
|                                                                        |             | MCWBR  | 9.1   | 9.8   | 9.5   | 8.7   | 7.9   | 6.4   | 4.8   | 4.5   | 5.1   | 5.6   | 6.3   | 8.2   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.591 | 0.548 | 0.557 | 0.626 | 0.699 | 0.699 | 0.601 | 0.570 | 0.580 | 0.593 | 0.651 | 0.651 |
|                                                                        | taud        |        | 1.678 | 1.811 | 1.825 | 1.739 | 1.648 | 1.669 | 1.892 | 1.991 | 1.942 | 1.846 | 1.648 | 1.619 |
|                                                                        | Ebn at Noon |        | 205   | 232   | 239   | 226   | 210   | 209   | 230   | 237   | 231   | 218   | 188   | 182   |
|                                                                        | Edh at Noon |        | 69    | 64    | 67    | 75    | 81    | 79    | 63    | 57    | 59    | 62    | 71    | 71    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1052  | 1428  | 1787  | 1966  | 1903  | 1636  | 1398  | 1460  | 1425  | 1407  | 1245  | 1020  |
|                                                                        | RadStd      |        | 95    | 89    | 69    | 100   | 120   | 114   | 101   | 85    | 83    | 115   | 66    | 102   |

**Historical Trends**

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) Station Only          | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page